Turtle Games Analysis – Report

Analyst: XXX

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Background/context of the business

Turtle Games is a game manufacturer and global retailer of books, board games, video games, and toys. The company collects data from sales and customer reviews.

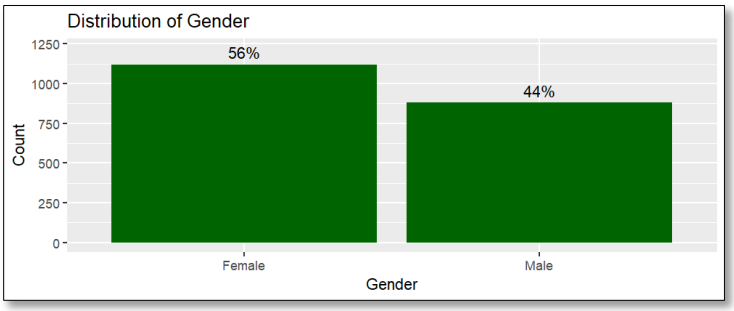
Its **business objective** is to improve overall sales performance by investigating customer trends:

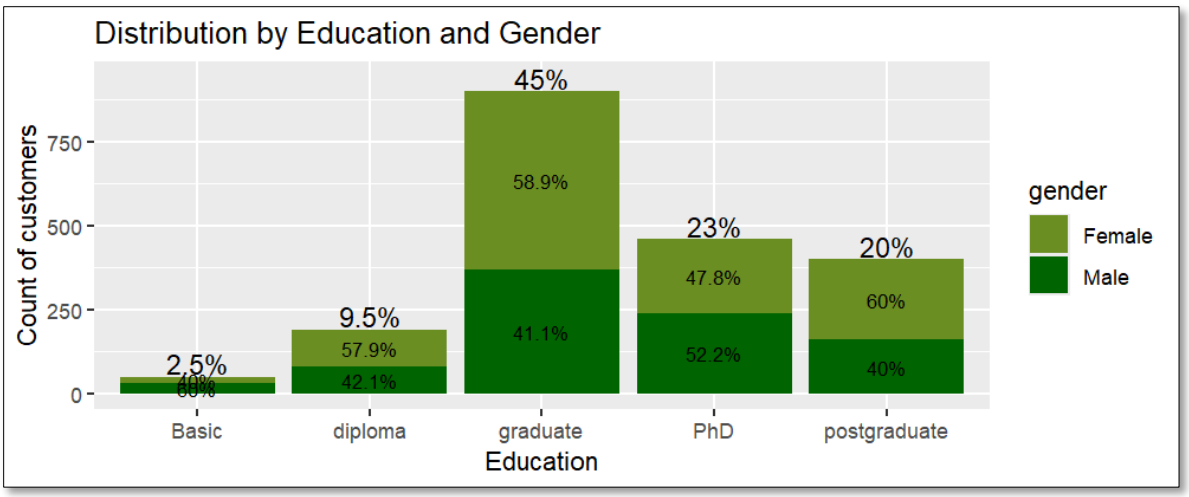
- How do customers engage with and accumulate **loyalty points**?
- How can customers be segmented into groups? **Which groups can be targeted** by marketing department?
- How can **text data** (e.g. social data such as customer reviews) be used to inform marketing campaigns and make improvements to the business?
- Do descriptive statistics provide insights into the suitability of the **loyalty points data to create predictive models**?

Analytical approach and visualisation

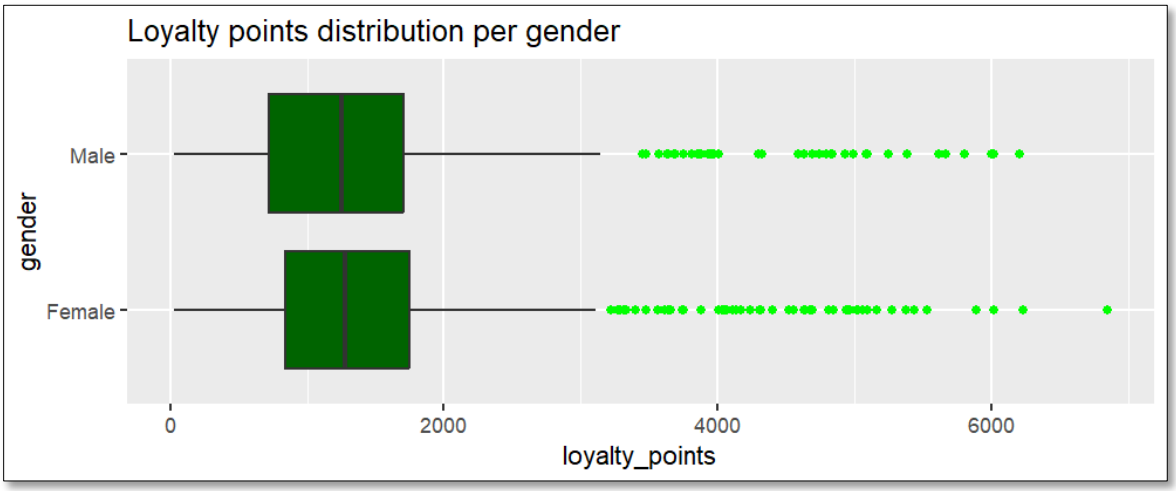
Model definition 1 – Linear regression:

1. **Loaded** the turtle_reviews.csv file, created a DataFrame 'reviews' and sense-checked data. No missing values identified.
2. **Cleaned** the DataFrame by dropping unnecessary columns and renaming headings. Age bins were created as a new column "age_group"
3. Performed **exploratory analysis** of the dataset:

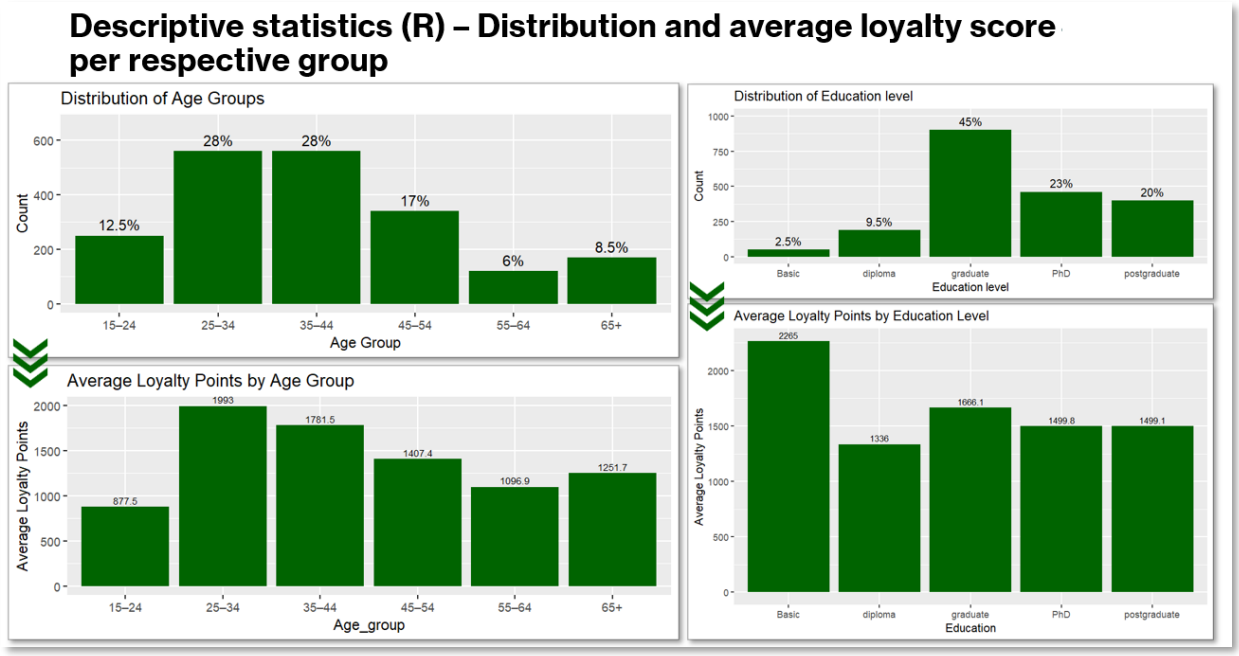




* Graduates are the biggest customer group followed by PhD and Postgraduates



No significant difference in loyalty score between males and females

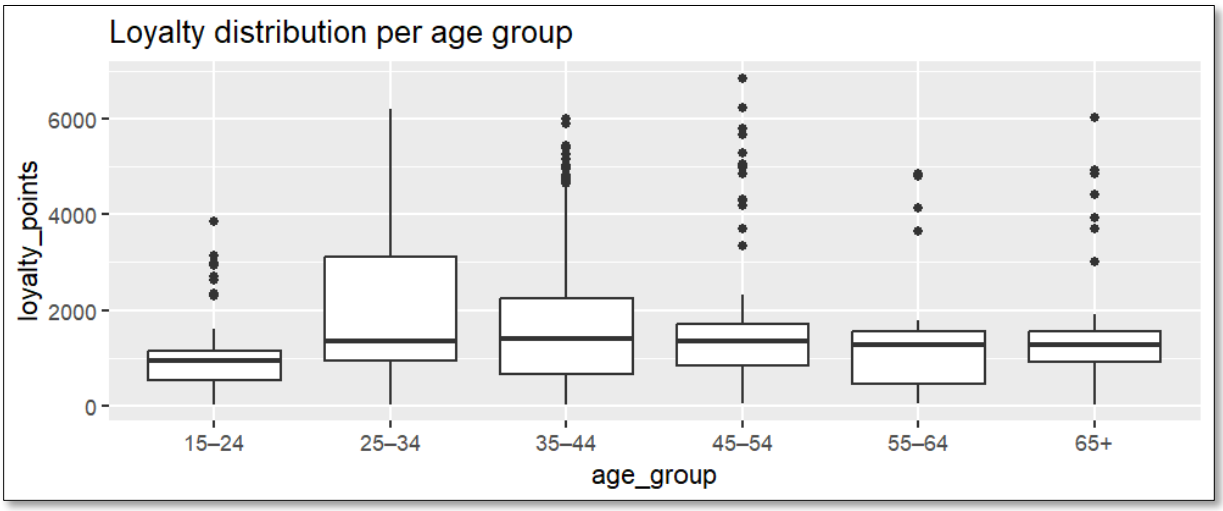


The most represented age groups have the highest average loyalty points.

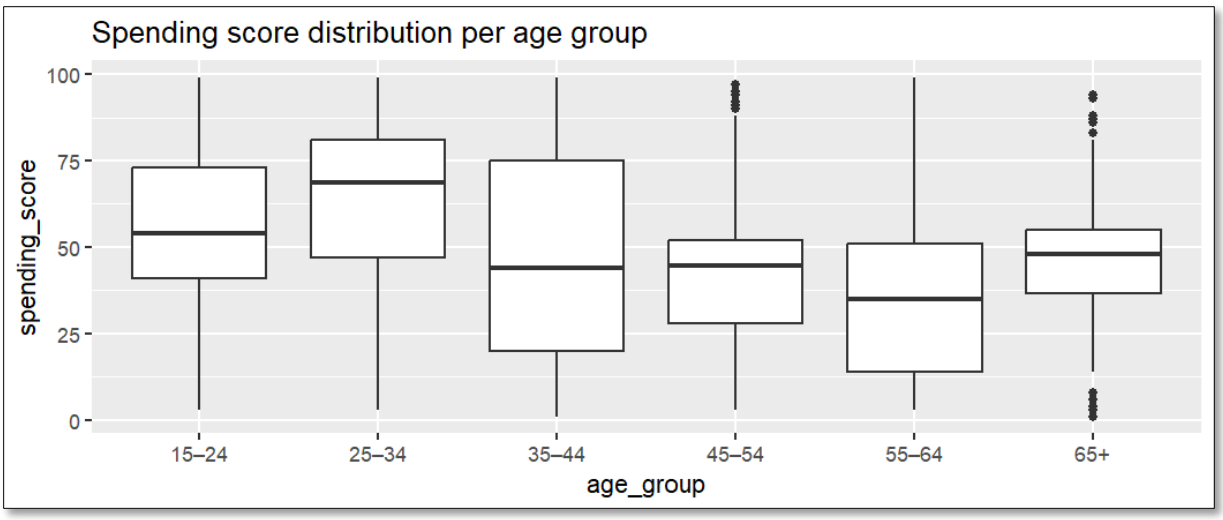
Although the basic education group is the smallest (n=50), it shows the highest average loyalty

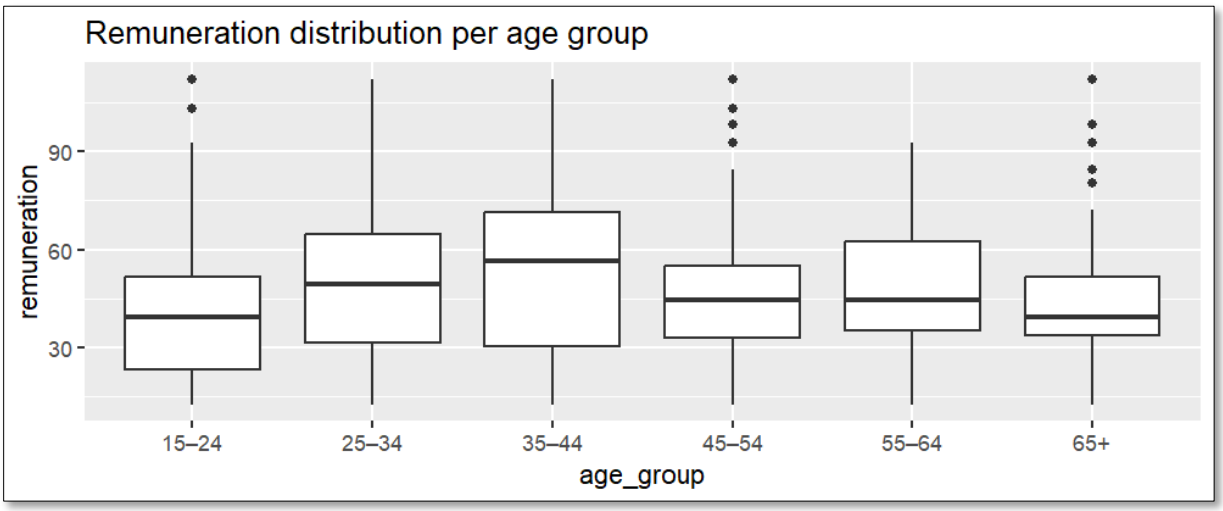
→ explore further

AGE:

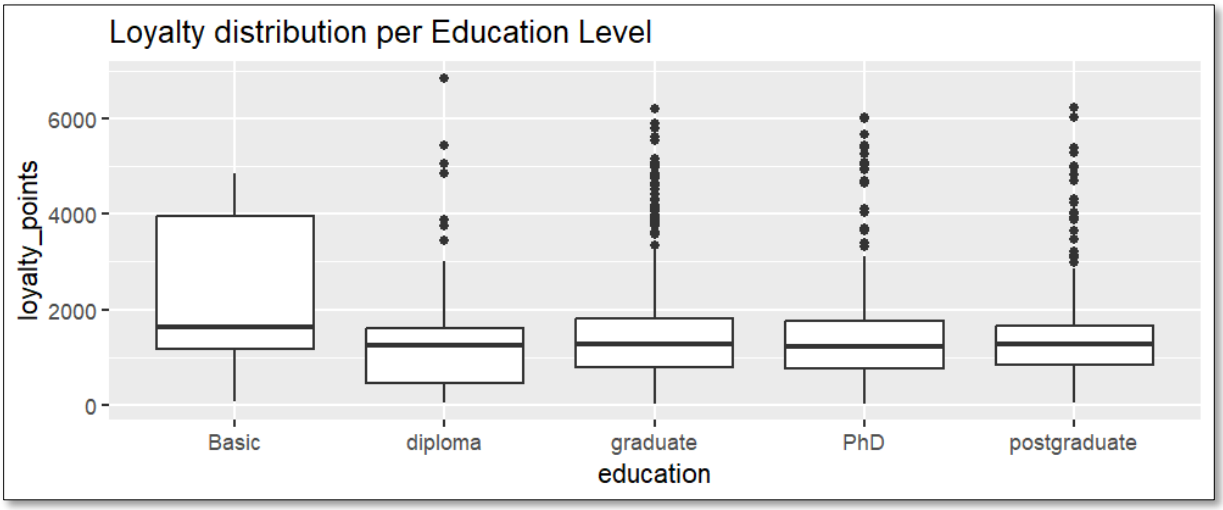


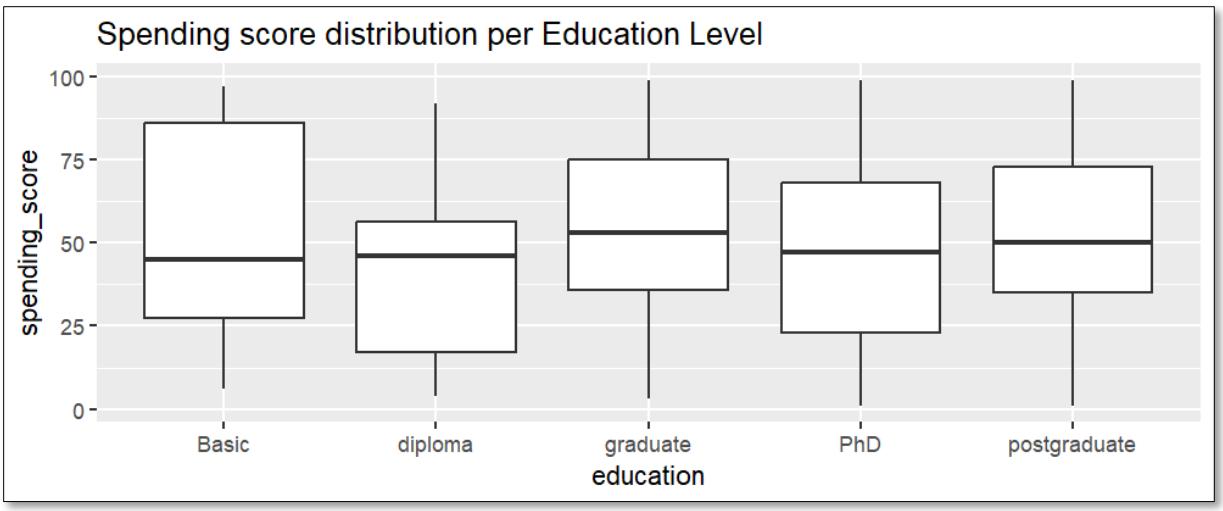
Age group 25-34 slightly more variance in loyalty, most frequent loyalty score is almost the same across age groups.



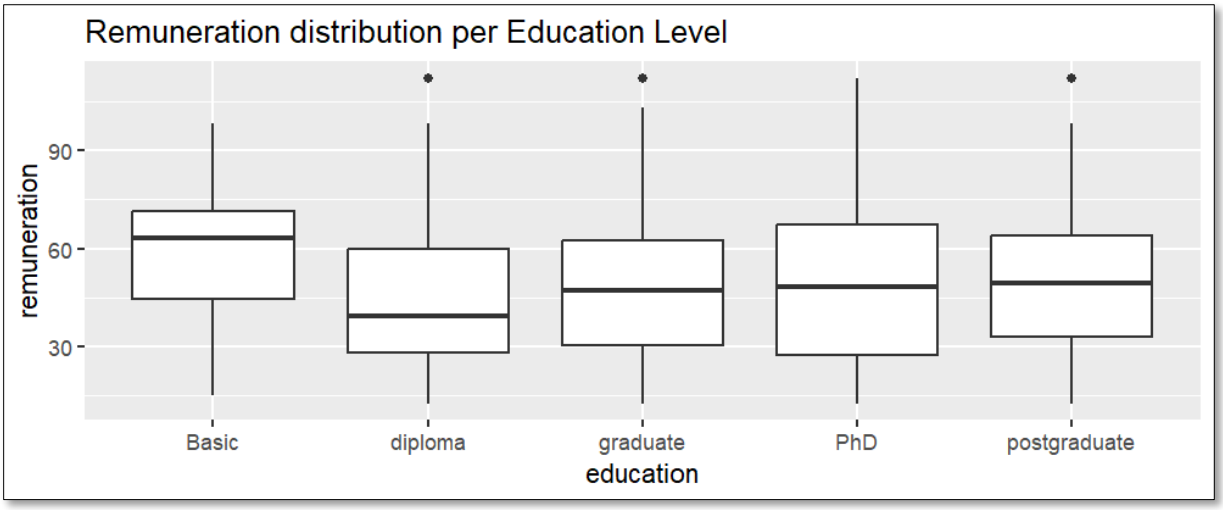


EDUCATION LEVEL:





Spending score of customers with basic education is slightly below overall spending average (50k).



Remuneration in basic education group is on average the highest. Diploma again on the lower end of spectrum.

- Basic education level has the highest most frequent loyalty score but represents only 2,5% of all customers (50 customers).
- Diploma has the lowest most common loyalty score with average spending and low income-> 9,5% customer share
- PhD and postgraduate account for 43% of customer pool → moderate spending and income
- Deep dive into basic education level:

```
> # Check descriptive stats for remuneration (filtered for basic level education)
> df_edu_basic %>%
+     summarise(
+     mean = mean(remuneration),
+     median = median(remuneration),
+     min = min(remuneration),
+     Q1 = quantile(remuneration, 0.25),
+     Q3 = quantile(remuneration, 0.75),
+     sd = sd(remuneration))
     mean median   min    Q1    Q3      sd
1 59.5812  63.14 14.76 44.28 71.34 19.5215
```

Comparing with statistics for the whole dataset:

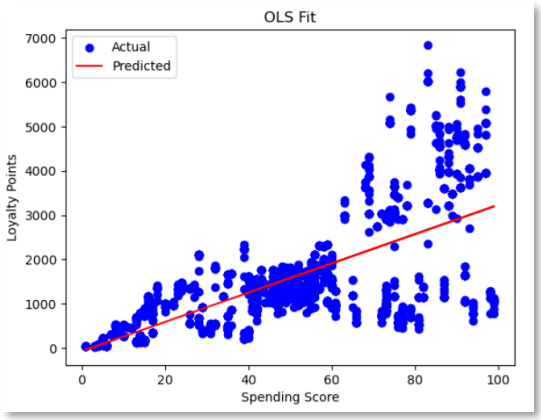
	age	remuneration (k£)	spending_score (1-100)	loyalty_points	product
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	39.495000	48.079060	50.000000	1578.032000	4320.521500
std	13.573212	23.123984	26.094702	1283.239705	3148.938839
min	17.000000	12.300000	1.000000	25.000000	107.000000
25%	29.000000	30.340000	32.000000	772.000000	1589.250000
50%	38.000000	47.150000	50.000000	1276.000000	3624.000000
75%	49.000000	63.960000	73.000000	1751.250000	6654.000000
max	72.000000	112.340000	99.000000	6847.000000	11086.000000

Customers with basic education have above-average remuneration and average spending - key loyalty factors (sample size: 50). Expanding marketing reach in this segment could grow the customer base.

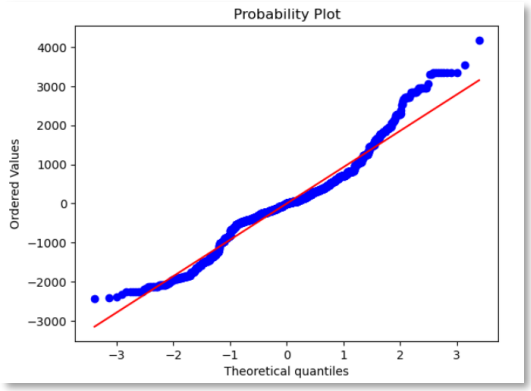
→ Perform **simple and multiple linear regression** to evaluate possible linear relationships between loyalty points and spending score / remuneration / age.

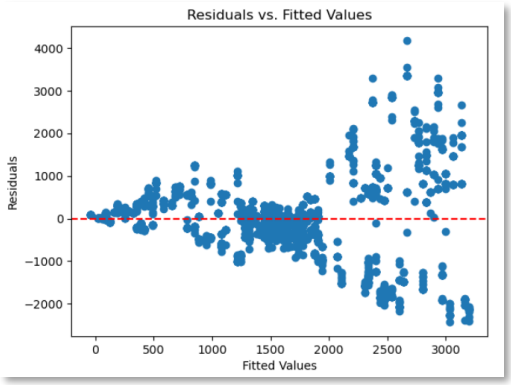
Simple linear regression

- The model with **'spending_score'** (though significantly relevant) explained only ca 45% of the total loyalty variance.

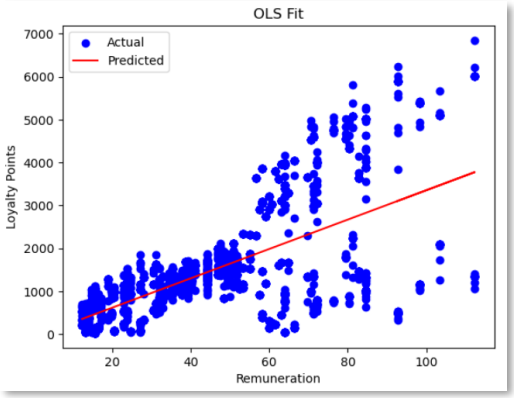


Residuals were roughly normal (see Q-Q plot below) and showed non-constant variance (heteroscedasticity).

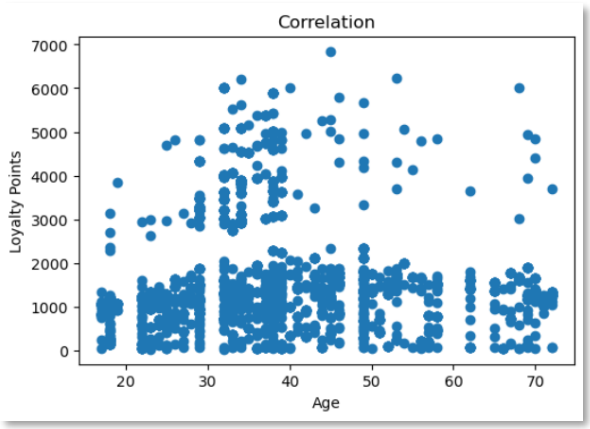




- The model with **'remuneration'** variable (though significantly relevant) explained only 38% of the total loyalty_point variance



- Variable '**age'** is not in linear relationship with loyalty (not suitable for linear regression)



Multiple linear regression

- model with independent variables 'remuneration' and 'spending_score' was trained with 70% of the data. Final model version including entire data set resulted in a **better fit than simple linear regression explaining 83% of loyalty variance.**

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.827
Model:                            OLS   Adj. R-squared:                  0.827
Method:                 Least Squares   F-statistic:                     2147.
Date:                Wed, 24 Sep 2025   Prob (F-statistic):               0.00
Time:                        10:10:44   Log-Likelihood:                -15398.
No. Observations:                2000   AIC:                         3.080e+04
Df Residuals:                    1997   BIC:                         3.082e+04
Df Model:                           2
Covariance Type:                  HC3
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept  -1700.3051     50.621    -33.589      0.000   -1799.521   -1601.090
x[0]          32.8927      0.635     51.821      0.000      31.649      34.137
x[1]          33.9795      0.790     43.032      0.000      32.432      35.527
==============================================================================
Omnibus:                        4.723   Durbin-Watson:                   3.477
Prob(Omnibus):                  0.094   Jarque-Bera (JB):                4.650
Skew:                           0.103   Prob(JB):                       0.0978
Kurtosis:                       3.115   Cond. No.                         220.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

*final version model considering entire dataset and accounting for heteroscedasticity

Error metrics:

```
MAE: 414.83
MSE: 284879.79
RMSE: 533.74
R² Score: 0.83
```

- By increasing spending score by 1, loyalty points increase by ca 33
- By increasing remuneration by 1, loyalty points increase by ca 34
- MAE suggests model's predictions are on average off by 415 loyalty points
- Moderate error of RSME = 533 is relative to standard error mean (1,578) – acceptable
- Data marginally right skewed, outliers have plausible values → decision to retain outlier data in the dataset

```
Skew:                           0.103
Kurtosis:                       3.115
```

- No multicollinearity detected

	VIF Factor	features
0	8.9	const
1	1.0	spending_score
2	1.0	remuneration

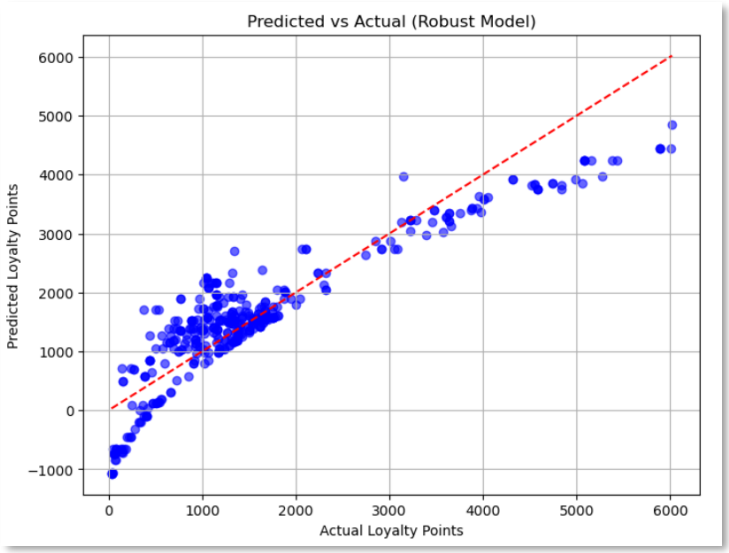
- Heteroscedasticity was accounted for with introduction of cov_type='HC3' -> model therewith more robust

Before HC3: After HC3:

t
-42.950
63.947
59.838

z
-33.589
51.821
43.032

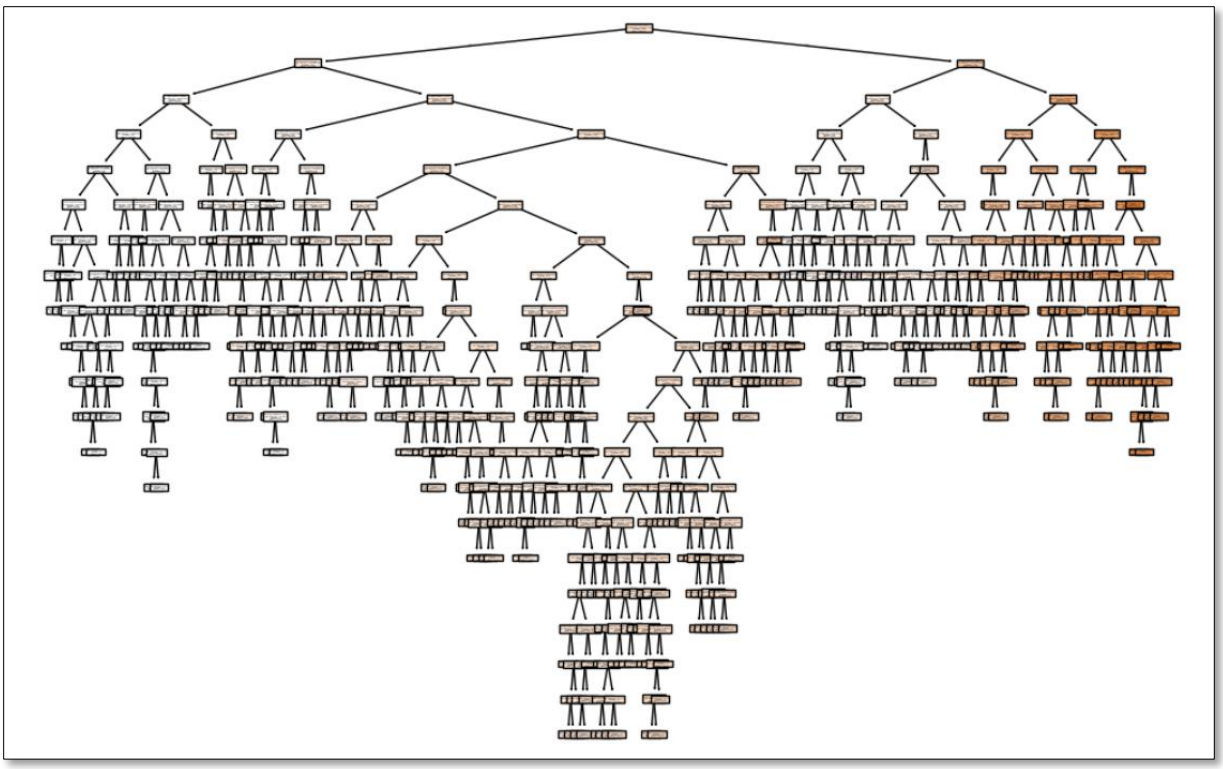
Visualisation of the model:



Model definition 2 – Decision Trees:

Before creating a decision tree, categorical variables were converted into dummy variables:

```
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 12 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   age                     2000 non-null   int64
 1   remuneration            2000 non-null   float64
 2   spending_score          2000 non-null   int64
 3   loyalty_points          2000 non-null   int64
 4   product                 2000 non-null   int64
 5   review                  2000 non-null   object
 6   summary                 2000 non-null   object
 7   gender_Male             2000 non-null   bool
 8   education_PhD           2000 non-null   bool
 9   education_diploma       2000 non-null   bool
 10  education_graduate      2000 non-null   bool
 11  education_postgraduate  2000 non-null   bool
dtypes: bool(5), float64(1), int64(4), object(2)
```



Full decision tree considering all variables in the clean dataset explains 99% of the variance but looks overfitted.

```
Mean Absolute Error:  44.33
Mean Squared Error:  14360.59
Root Mean Squared Error: 119.84
R² Score: 0.99
```

→ too complex, need to deduce non-significant variables and prune tree

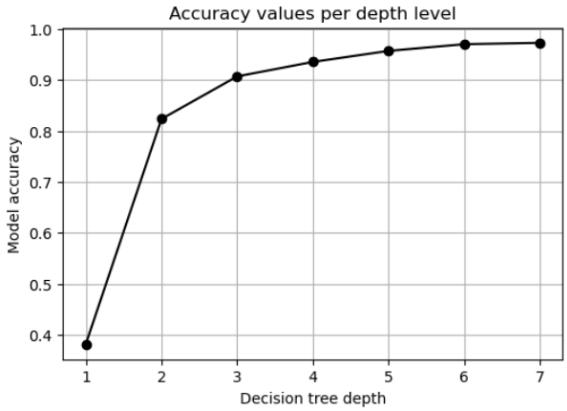
Decision to focus only on remuneration and spending score variables as per feature importance evaluation:

```
remuneration              0.543051
spending_score            0.439818
age                       0.015124
gender_Male               0.000781
education_graduate        0.000401
education_diploma         0.000396
education_PhD             0.000263
education_postgraduate    0.000166
dtype: float64
```

→ conclusive with results of multiple linear regression model

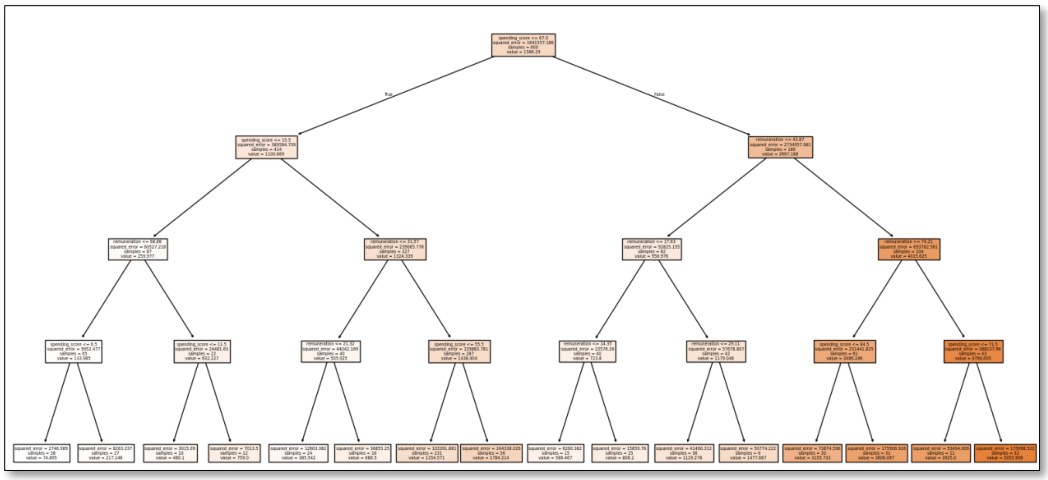
```
Full tree excl. non-significant x variables:
Mean Absolute Error:  90.92
Mean Squared Error:  32391.63
Root Mean Squared Error: 179.98
R² Score: 0.98
```

To determine the suitable pruning level, following metrics were used:



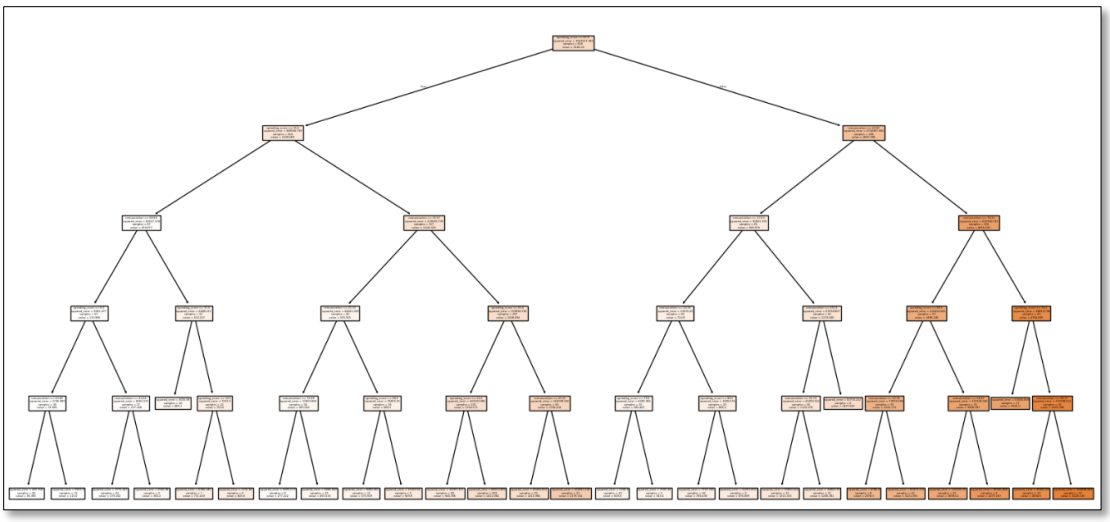
- Max depth = 4:

```
Prunned tree (depth = 4) excl. non-significant x variables:
Mean Absolute Error:  231.56
Mean Squared Error:  106198.63
Root Mean Squared Error: 325.88
R² Score: 0.94
```



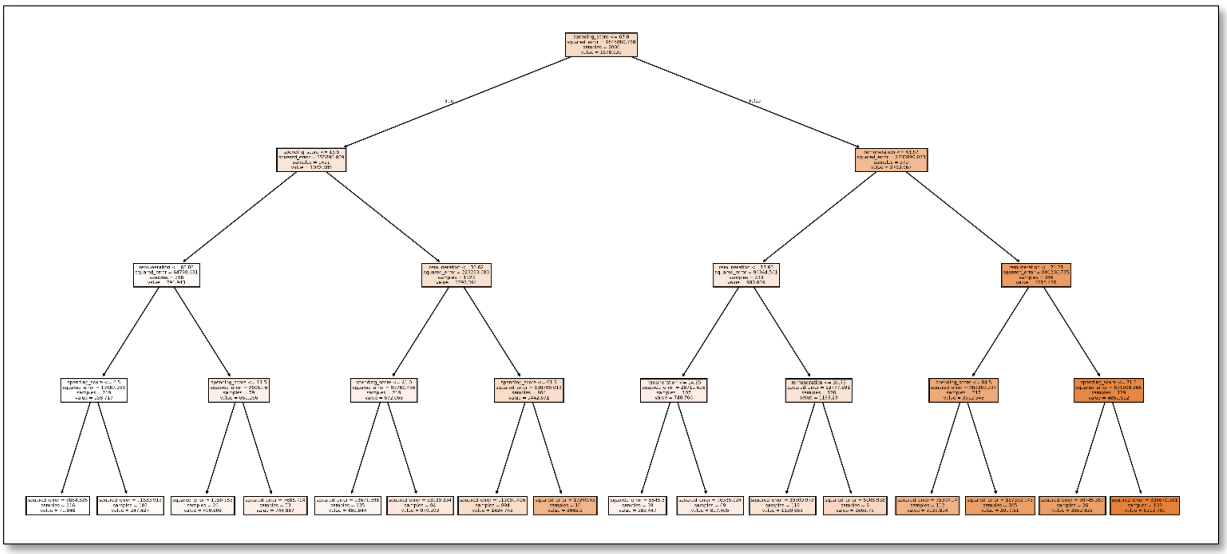
- Max depth = 5:

```
Prunned tree (depth = 5) excl. non-significant x variables:
Mean Absolute Error:  188.74
Mean Squared Error:  71833.88
Root Mean Squared Error: 268.02
R² Score: 0.96
```

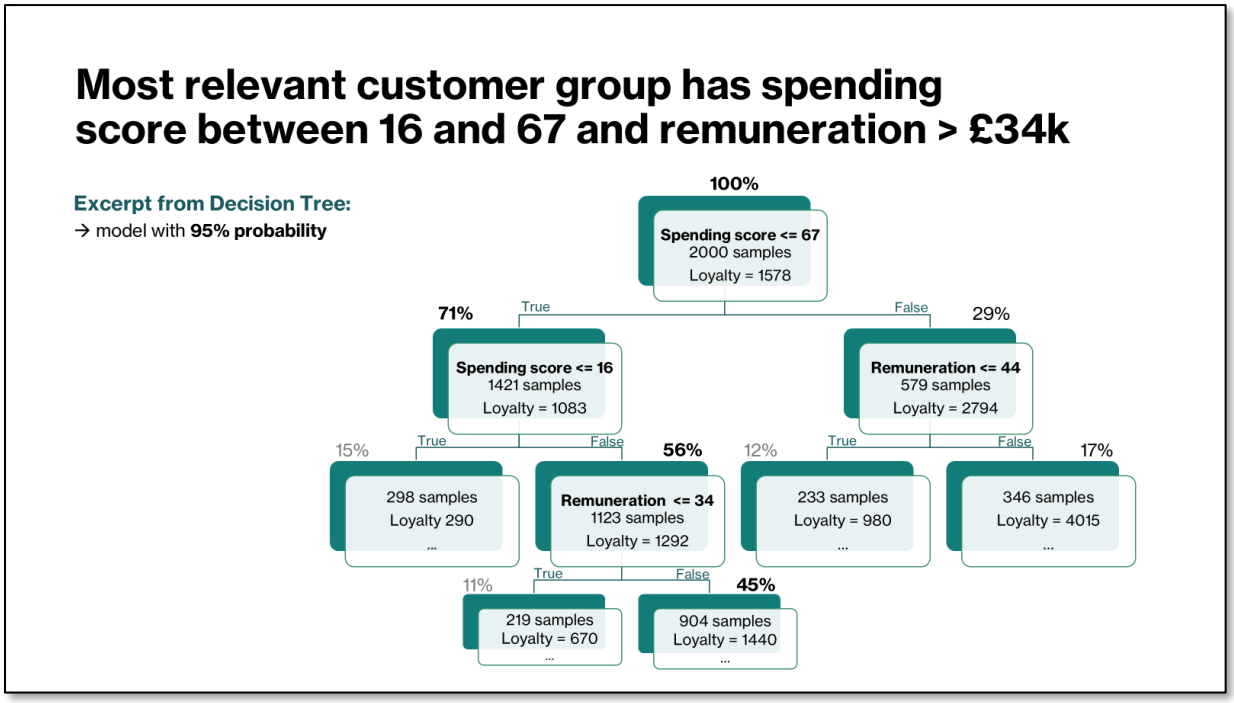


While max depth = 5 compared to depth of 4 slightly improves accuracy and lowers error metrics, the increased complexity reduces interpretability and offers limited business insights → **opting for decision tree model with max depth = 4:**

```
Final model (depth = 4 incl. remuneration & spending_score):
Tree depth = 4
Number of leaves = 16
Mean Absolute Error:  210.79
Mean Squared Error:  89775.49
Root Mean Squared Error: 299.63
R² Score: 0.95
```



Following insights gained:



- **45%: Biggest customer group with spending score between 16 and 67 and remuneration higher than £34k → moderate loyalty score** (overall loyalty-mean = ~1.580)
- **17%:** Spending score above 67 and remuneration above £44k → very high loyalty score
- **15%:** Spending score lower than 16 → low loyalty
- **12%: Spending score above 67 and remuneration below £44k → lower loyalty than those with spending score below 67**
- **11%:** Spending score above 67 but remuneration < £34k → moderate loyalty

Remuneration appears to have a stronger influence on loyalty than spending, although both contribute to increasing loyalty score.

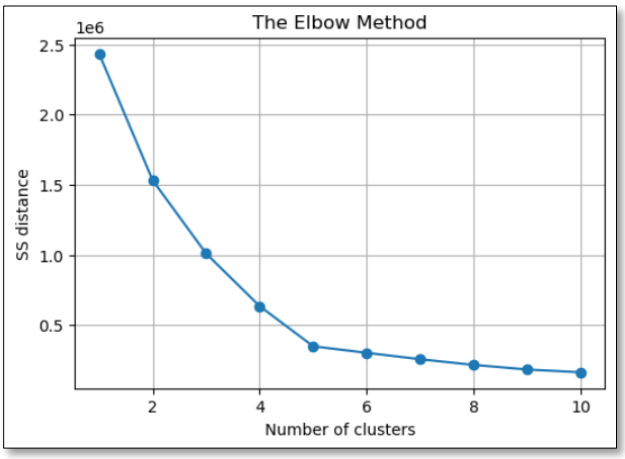
Results are conclusive with random forest regression model in terms of significant variables and relevant cut-off values (see Jupyter Notebook, end of section 2 “Decision tree analysis” for more detail).

Model definition 3 – Clustering

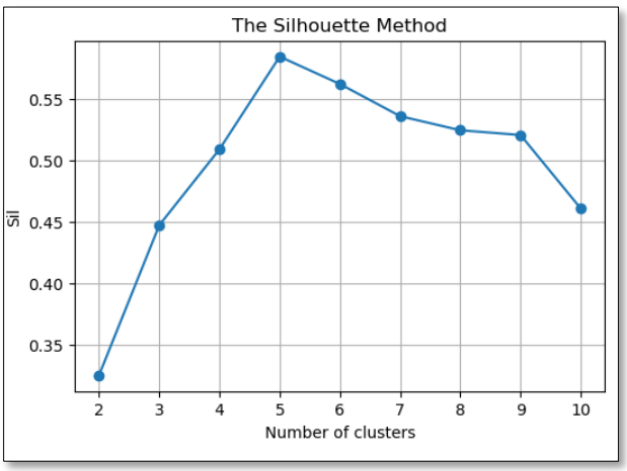
New DataFrame was created excluding irrelevant columns:

	remuneration	spending_score	loyalty_points
count	2000.000000	2000.000000	2000.000000
mean	48.079060	50.000000	1578.032000
std	23.123984	26.094702	1283.239705
min	12.300000	1.000000	25.000000
25%	30.340000	32.000000	772.000000
50%	47.150000	50.000000	1276.000000
75%	63.960000	73.000000	1751.250000
max	112.340000	99.000000	6847.000000

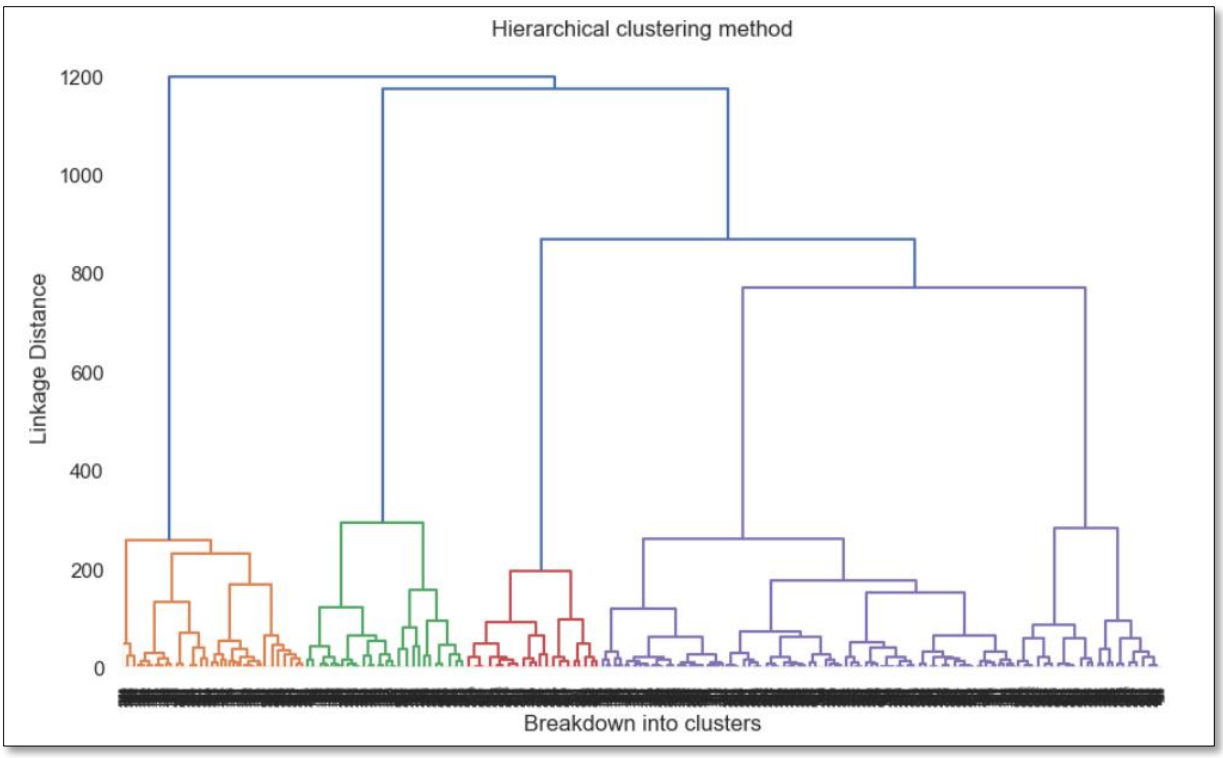
Applied elbow, silhouette and hierarchical clustering methods to estimate optimal number of clusters:



- Drop in ss (sum of squared) distance starts flattening after 5 clusters



- Highest silhouette at 5 clusters



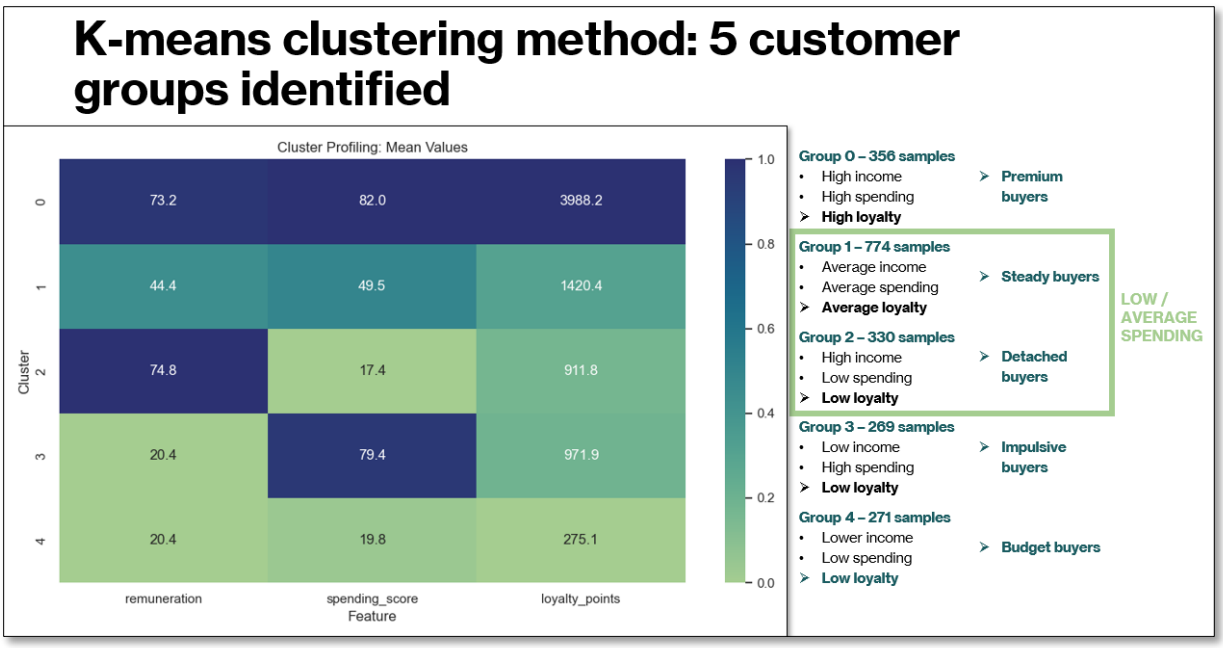
- Distance between merged clusters still significantly reduces with 5 clusters

→ K-means model altering number of clusters (1-5) was run to verify the best fit (see Jupyter notebook, section 3.4 for respective visualisations)

Model with 5 cluster chosen as the best fit based on elbow, silhouette and hierarchical clustering methods as well as the interplay of group sizes and interpretability for the business as per k-means results:

K-means method:

Nr. of clusters	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
2	894 45%	1106 55%			
3	429 21%	1220 61%	351 18%		
4	356 18%	1038 52%	337 17%	269 13%	
5	356 18%	774 39%	330 17%	269 13%	271 14%



Specific characteristics within clusters:

age_group cluster	15–24	25–34	35–44	45–54	55–64	65+
0	0.02	0.54	0.37	0.04	0.01	0.02
1	0.14	0.22	0.18	0.24	0.06	0.15
2	0.09	0.13	0.44	0.19	0.12	0.03
3	0.26	0.44	0.19	0.06	0.02	0.03
4	0.11	0.13	0.34	0.22	0.09	0.12

education cluster	Basic	PhD	diploma	graduate	postgraduate
0	0.05	0.17	0.08	0.53	0.17
1	0.01	0.21	0.10	0.45	0.23
2	0.05	0.30	0.08	0.35	0.22
3	0.00	0.26	0.04	0.50	0.20
4	0.01	0.24	0.19	0.42	0.14

gender cluster	Female	Male
0	0.55	0.45
1	0.58	0.42
2	0.48	0.52
3	0.56	0.44
4	0.63	0.37

Customer groups:

Priority A -High Impact

Group 2 – Detached Buyers (n=330)
Traits: High income, low spending, low loyalty // Graduates, postgraduates, PhD // mostly aged 35-44
Strategy: Re-engage with emotional value

- Highlight brand values, sustainability, or social impact
- Offer experiential rewards (e.g., events, memberships)
- Use storytelling in campaigns to build emotional connection
- Test premium product trials or concierge services

Group 1 – Steady Buyers (n=774)
Traits: Average income, average spending, average loyalty // mostly graduates // aged 25-54 // 60% females
Strategy: Nurture and grow

- Use personalized recommendations based on past purchases
- Offer limited-time bundles or cross-sell opportunities
- Send educational content to deepen brand connection

Priority B - Lower impact

Group 0 – Premium Buyers (n=356)
Traits: High income, high spending, high loyalty
Strategy: Retain Strategy: Retain (reinforce exclusivity through VIP access and special member-only benefits)

Group 4 – Budget Buyers (n=271)
Traits: Lower income, low spending, low loyalty
Strategy: Maintain at Low Cost (via automated discounts and low-effort promotions)

Group 3 – Impulsive Buyers (n=269)
Traits: Low income, high spending, low loyalty
Strategy: Stabilise (through simple, consistent-purchase rewards)

Analysis in R

Clean dataset was loaded:

```
> str(df)
'data.frame':   2000 obs. of  9 variables:
 $ gender        : chr  "Male" "Male" "Female" "Female" ...
 $ age           : int  18 23 22 25 33 24 37 25 66 32 ...
 $ remuneration  : num  12.3 12.3 13.1 13.1 13.9 ...
 $ spending_score: int  39 81 6 77 40 76 6 94 3 72 ...
 $ loyalty_points: int  210 524 40 562 366 573 61 772 31 724 ...
 $ education     : chr  "graduate" "graduate" "graduate" "graduate" ...
 $ product       : int  453 466 254 263 291 977 979 1012 1031 1459 ...
 $ review        : chr  "When it comes to a DM's screen, the space on the screen itself is at an ab
solute premium. The fact that 50% of "| __truncated__ "An Open Letter to GaleForce9*:\n\nYour unpai
nted miniatures are very not bad. Your spell cards are great. Your "| __truncated__ "Nice art, nice
printing.  Why two panels are filled with a general Forgotten Realms map is beyond me.  Most of "|
__truncated__ "Amazing buy! Bought it as a gift for our new dm and it's perfect!" ...
 $ summary       : chr  "The fact that 50% of this space is wasted on art (and not terribly informa
tive or needed art ..." "Another worthless Dungeon Master's screen from GaleForce9" "pretty, but al
so pretty useless" "Five Stars" ...
```

```
> summary(df)
    gender               age          remuneration     spending_score loyalty_points
 Length:2000        Min.   :17.00   Min.   : 12.30   Min.   : 1     Min.   :  25
 Class :character   1st Qu.:29.00   1st Qu.: 30.34   1st Qu.:32     1st Qu.: 772
 Mode  :character   Median :38.00   Median : 47.15   Median :50     Median :1276
                    Mean   :39.49   Mean   : 48.08   Mean   :50     Mean   :1578
                    3rd Qu.:49.00   3rd Qu.: 63.96   3rd Qu.:73     3rd Qu.:1751
                    Max.   :72.00   Max.   :112.34   Max.   :99     Max.   :6847
  education            product          review            summary
 Length:2000        Min.   :  107   Length:2000        Length:2000
 Class :character   1st Qu.: 1589   Class :character   Class :character
 Mode  :character   Median : 3624   Mode  :character   Mode  :character
                    Mean   : 4321
                    3rd Qu.: 6654
                    Max.   :11086
```

Multiple linear regression in R automatically converts categorical to dummy variables.

The model was trained on 70% of the data.
'Education' variables were removed from the model since not statistically significant.

Based on the metrics of adjusted R^2 and RMSE as well as model complexity, best fit resulted to be model considering remuneration, spending score and age.

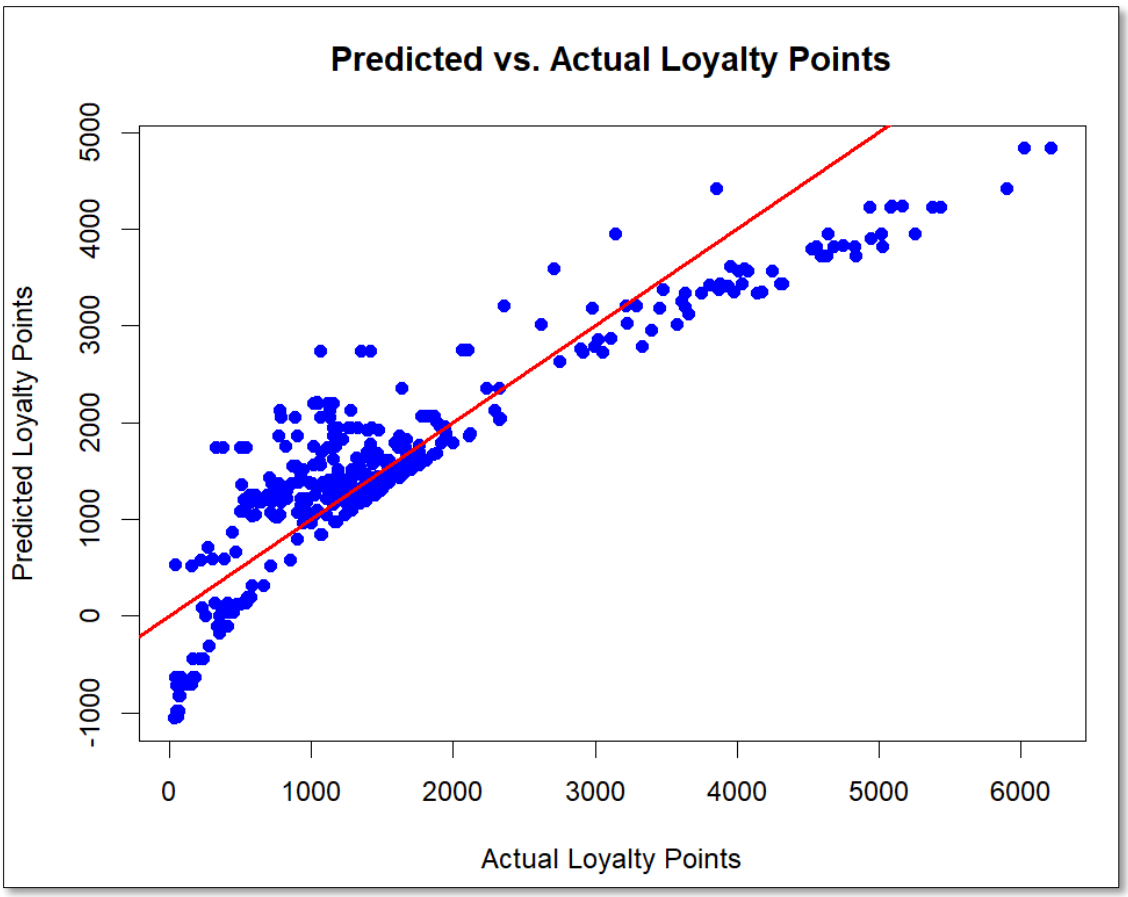
```
Call:
lm(formula = loyalty_points ~ remuneration + spending_score +
    age, data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1839.01  -340.84    15.36   275.30  1912.26

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)    -2161.3231    60.7492  -35.58   <2e-16 ***
remuneration      34.1435     0.5963   57.26   <2e-16 ***
spending_score    33.4666     0.5322   62.89   <2e-16 ***
age               10.7257     1.0053   10.67   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 503.2 on 1396 degrees of freedom
Multiple R-squared:  0.8389,    Adjusted R-squared:  0.8386
F-statistic:  2423 on 3 and 1396 DF,  p-value: < 2.2e-16

RMSE: 538.85
>     cat("Test R-squared:", round(r_squared_test, 4), "\n")
Test R-squared: 0.8411
```

Model with only remuneration and spending score had slightly worse accuracy metrics which corresponded to MLR analysis performed in Python.

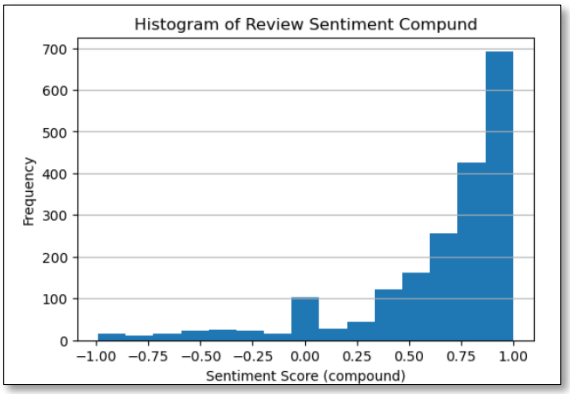
Since the age variable is non-linear, meeting the assumptions of MLR would require transforming the data. Considering limited gains in accuracy, age was disregarded from the MLR analysis.

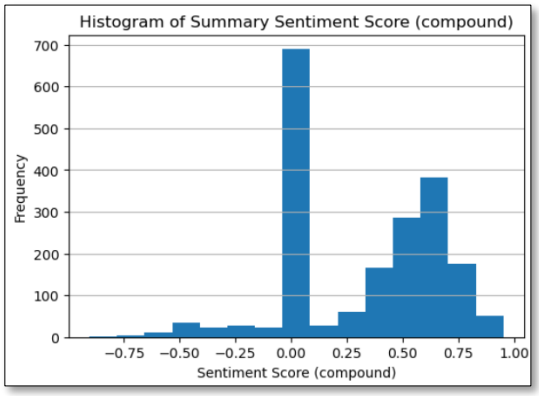
```
Call:
lm(formula = loyalty_points ~ remuneration + spending_score,
    data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1526.36  -362.56    43.69   277.94  2010.36

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)    -1680.2998    42.3258  -39.70   <2e-16 ***
remuneration      34.1378     0.6199   55.07   <2e-16 ***
spending_score    32.3120     0.5417   59.65   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 523.1 on 1397 degrees of freedom
Multiple R-squared:  0.8258,    Adjusted R-squared:  0.8255
F-statistic:  3311 on 2 and 1397 DF,  p-value: < 2.2e-16

RMSE: 559.74
>     cat("Test R-squared:", round(r_squared_test, 4), "\n")
Test R-squared: 0.8285
```

```
vif(model)
remuneration spending_score
    1.000021       1.000021
```



Model validation 4 – Text / Sentiment analysis





Compound sentiment score considering polarity and subjectivity for the review section shows a clear positive trend. Most common sentiment scoring for 'Summary' section was neutral. Negative comments are in minority.

→ Based on neg comments, biggest resentment is the quality of the product (complaints on paper quality and design)

→ Sentiment scores for comments (reviews or summaries) are sometimes questionably assigned resulting in misrepresented sentiment trend.

Excerpts:

Customer review text analysis shows strong positive sentiment, though scoring is inconsistently assigned

Excerpt from Top 15 negative comments:

Comment ID	Review Text	Score
REVIEW #280	I bought this thinking it would be really fun but I was disappointed its really messy and it isnt nearly as easy as it seems also the glue is useless for a 9 year old the instructions are very difficult	-0.9520
REVIEW #1053	for those just getting started in the dnd world or those who havent played dnd in years if not decades this is a good way to get learn, reacclimate yourself to the games played an adventure with a friend on the first day while the amount of pieces tokens tiles was a little overwhelming at first got accustomed to the game pretty quickly and had quite a bit of fun my only con about it is that ashaldaronthe figure seemed well it had great detail on it but the **assembling in factory seemed bad or poor there is a slight gap between the two sections** and what appears to be a slight overflow of glue that was used i guess i kind of expected a one piece mold or if not a bit more care taken when assembling i do plan on painting these figures at some point and im not sure whether this is normal or not as well as whether it will be possible to really paint this particular figure without it looking off due to the gapglue problem perhaps i got a bad piece ill probably end up talking to customer service about this issue and see about getting **a replacement** that was assembled properly	**-0.9539**
REVIEW #889	the game is amazing the new version is not cardboard is biodegradable but a greener version could have been produced with better game pieces i guess avalon hill has beaten us all at its own board game they no longer need to cater to the consumer because people will buy even **though this version is terrible in comparison to the 1999 edition** disappointing but not surprising	-0.8984
SUMMARY #1453	Adds a needed amount of tension in a tense fantasy world	-0.5719
SUMMARY # 172	Horrible Nothing more to say Would give zero stars	**-0.5423**
SUMMARY #1508	Very cute but poor quality stickers	-0.5266

Despite its limitations, NLP enables scalable insight extraction from lengthy customer comments

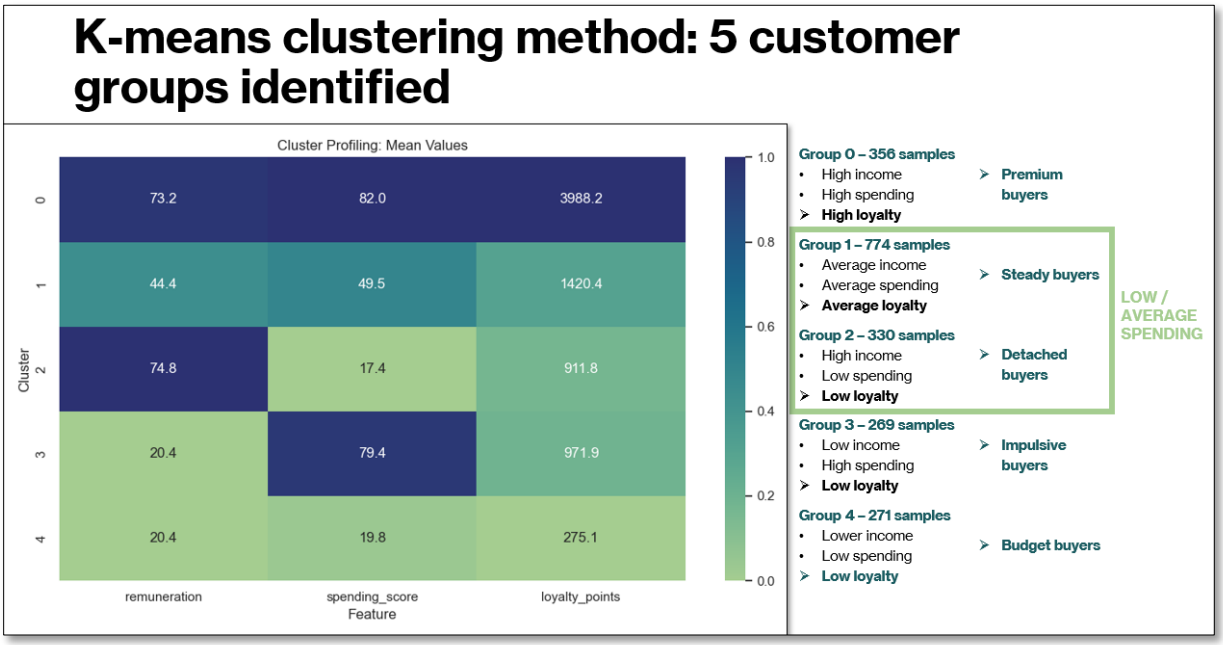
Excerpt from Top 15 positive comments (for illustrative purposes):

1105	disclaimer bought this from a local store paid list value but supporting local game stores helps keep them in business and its a rough market to keep a gaming store running also i do not own nor have i playedintrpgs datahookproductlinklinked classlinknormal hrefungeonsdragonsactersworldfbookandgamedp/0786955770ref=m_cr_arp_d_mw_tc&keyBlu&qid=1 dragons castle ravenloft board games the first dd adventures board game regardless dungeons dragons wrath of ashardalon board game rule is a fantastic value at 65 list you get 42 plastic unpainted minis 1 scale you get a nice set of cardboard interlocking dungeon tiles and you get a fun very random board game playable either by yourself or with up to 4 friends what it is rule is in gamer terms a roguelike in tabletop form in layman terms rose is a dungeon crawler you start at the entrance and have a specific goal rescue the villagers survive the dungeon destroy the altar slay the dragon the adventures are built randomly based on shuffling the tiles and they are played individually as you explore you can go for a massive linear dungeon based on a little luck and careful planning or you can attempt to condense yourself in a larger swarm interconnected tunnels some of the adventures feature chambers special larger rooms that are all placed at once and filled with enemies and a specific goal that the team must complete to win the adventure its a great way to introduce someone into tabletop roleplaying gaming its the perfect medium between a board game and an actual pen and paper rpg its a fun quick game to run a quick adventure for some friends taking anywhere between 20 minutes and an hour per adventure what it isn't a deeper roleplaying game with a full interconnected series of quests over a grand campaign a completely accurate representation of dth edition dungeons dragons rules overly complicated as most roleplaying games will seem to be for most people whats in the box 42 plastic heroes and monsters 13 sheets of interlocking cardstock dungeon tiles 200 encounter and treasure cards rulebook scenario book 20cided die miniatures the game includes 42 minis including 7 villain 5 hero and 30 monsters the minis are of nice quality theyre clearly just mass produced but they put some effort into modeling them just not so much into trimming the leftover plastic in places some of the avoids will be bent from being in the box and theyre unpainted however i dont mind the unpainted state and i plan on using them during dd encounters im driving at a local store the adventadon figurine is very detailed my only and the only real gripe is that none of them are painted 43 cardstock dungeon tiles theyre one of the nicest parts of the set theyre very high quality the art is nice and the interlocking design can be beat purchasers of dd dungeon tiles will know how hard they are to keep together without tape these lock together like puzzle pieces wouldnt be hard to use them in a real dd game except for the fact that theyre somewhat limited and theyre not large open spaces there are a lot of tokens and no markers and other things all made out of the same cardstock including cards for the villain your big bad guys and the heroes 55 200 encounter monster treasure cards there are drawn during the different phases of the game there are 3 and they certainly add to the randomness of the game everything scales nicely so that you can attempt to run the party adventures solo and till still work nicely 55 rulebook the rulebook is the only real downfall ive seen of the box set it does an slightly job at conveying the basics but other questions i have are just nonexistent can i spend a heating surge without being dead just to heal the book itself has card errata on the back page which is crazy other than that its not a bad rulebook for being only 15 pages 355 scenario book adventure book this has 13 adventures to run using the set each of them varying based on how things shuffled what magic items get drawn and if theres a chamber what chamber card gets drawn the adventures essentially play themselves as you follow the turn orders theres very little fluff for each of the adventures but thats what make it dd you get to set your own motivation for the adventure your own background its enough to get you going and you let your imagination take it from there 455 20 cided die aka the d20 the iconic dd symbol you not it add a modifier and compare it to a set number thats pretty much it its just as nice as the dd red box dice or any cheesie dice ive bought but its just a d20 nothing special nice if you dont have any threw it in the bag with all your others 55 overall i would have liked to see more adventures than not to say more aren't coming the official ones available at time of review require castle ravenloft as well which i dont have yet theres only one solo adventure but i played through it and adventure 5 solo not sure if a single player could take on ashardalon himself but half the fun is seeing how well you can do i actually wouldnt mind running a full dd campaign with similar designs its a perfect format for a megadungeon delve its fast and its exciting due to the randomness im glad the cards aren't collectable theres 200 thats all there is all in all if youre into rogge or if youve ever wondered if a roguelike could work in a tabletop format it can and its called wrath of ashardalon edit i forgot to mention the initial setup punching out the cards takes a considerable amount of time like close to 20 minutes setup pel adventure doesnt take nearly as long maybe 20 minutes	0.0000
869	whenever i see this game on my shelf i get a disturbing visual of quarto big head from star trek deep space nine i then picture him playing tongo with a bunch of other flenergia game that deals with buying selling acquiring and other things youd normally see on the wall street floor dont ask me why i haven't had my coffee yet this morning acquire 35 players ages 12 average play time 60 to 90 minutes note while the box says that the game is for three to six players i dont see why a two player variant couldnt exist one doesnt already im fairly confident that two players with an imagination who are not concerned about following the rulebook to the letter could make do i also came across the older avalon hill versions rulebook and there is a section for special rules for two players feel free to experiment acquire plays a bit differently than the other games ive played the closest comparison i could possibly make to what ive already reviewed is airlines europa in both games youre buying stock and whoever has more of certain stocks colors often reaps the biggest reward when payouts occur more on that in a minute version components the version of acquire i purchased is the latest print of the game by wizards of the coast i need to get this pet peeve of mine out of the way before i begin the player tile trays are made of fairly cheap cardboard that you have to punch out and bend to make the pieces assemble in other words i had a fun time trying to assemble the player tile trays in most games ive purchased tile trays came assembled and were of much better quality some people may not mind the extra work and the game is fairly cheaper than your average eurotyle board game but cmonwould it have hurt profits that much to just make the components a little better also the player reference tables were part of the manual itself that i had to cut rear out for cod people like me the manual just doesnt look right now that pages look torn out of it i dont see why they couldnt have printed reference sheets separate of the manual previous versions of the game were much nicer looking and of better quality so much in fact that other reviewers recommend trying to find those older printed versions as opposed to this one i dont mind average to poor production values if the price is right but this is just pushing it game mechanics the game board is made of up of squares that form a grid going from 1a in the upper left hand corner all the way to 12 i in the lower right hand corner to the left of the grid are the available corporations that players can form and to the right is the game reference chart you'll find in the cutouts in the back of the manual in acquire turns are fairly simple players place a tile players buy up to three stocks and finally players pick up a new tile from the supply tiles correspond to the squares grid on the boardfor examplepoul place the 1a tile if you had it onto the 1a square on the board grid when a player places a tile next to an existing tile sitting by itself that player gets to form a corporation it doesnt become theirs mind you but they do get a free stock card for establishing it that player can choose any of the available corporationthough some grow differently than others some corporations are cheap to buy stocks for early on but dont pay out as much when they grow in size while others are expensive to buy stocks for early on but payout is big in the endgame players continue placing tiles forming growing corporations and buying stock cards until a tile is placed that causes two corporations to bump heads this is the real meat and potatoes of the game a few things happen when this occurs 1 the larger corporation acquires or eats up the smaller one if the purple corporation for example has six tiles and green has three tiles purple would acquire green the exception to the rule is if both corporations across the board are of the same size in size corporations can not merge 2 the player who owns the most stocks in the merging corporation gets a bonus payout 3 players who own stock in the merging corporation can sell on how big the corporation actually is the player who owns the most stock of the merging corporation receives the largest payout 3 players who own stock in the merging corporation can trade their stock for stock in the acquiring corporation or they can trade that stock on a 2 for 1 basis for stock of the company that acquired it they can keep their stock hoping the corporation will form again down the line the acquired company market gets put back to its place on the side of the board and can be formed again later on at the end of the game shareholder bonus amounts are given out to the companies that are on the board and their stocks are traded in for cash the winner is the player with the most money after that is all said and done the review all three kids joined me to form a four player game explaining the rules seemed as bad as i thought it could be i was concerned that the whole buying stocks and acquiring could overwhelm the eleven year old but ironically he ended up winning the game he randomly bought stock so that had to try to have more than everyone else which ended up working out to his advantage unbeknownst to him i focused on buying stocks for two corporations in the middle of the board and would have made a lot of money had my opponents acquired the dwarfed they ended up becoming safe and i didnt have much money left to buy stocks in other corporations that could still be acquired i received a nice bonus at the end of the game for all the stock i had but my son used the money he earned from his acquisitions to buy even more stock than what i had bought up cheaply in the beginning of the games he ended up getting the majority shareholder bonus anyway he earned so much from a particular acquisition that he just started buying out all of the stock cards of particular colors until there were none left it seems to pay to have money throughout the game and not to focus on one or two corporations unless you are sure they will pay out i was tempted to offer him a later bed time if that clip me ten grand but i thought better of it the game took us about two hours to playtwo in mind there was a learning curve and three kids of varying ages involved overall i really enjoyed the game acquire is simple enough to learn but deep enough that players can form strategies in order to outdo the other the random tile placement throughout the game by players keeps the replayability high i quickly forgot about the quality of the player tile trays when i started playing our scenario in my head on how certain corporations might play out on the board the kids were actively working out aloud which corporations they planned to grow and were constantly comparing how many stock cards they had of certain colors they were often on the edge of their seats when a merger occuredenvious to see how it would play out the older ones took it a step further and bought out stock in the growing corporations recognizing how much money they could make when it grows even bigger the younger one just bought what he could afford and random chance worked in his favor they all stated that they enjoyed playing the game i personally recommend acquire if you enjoy games with a financial theme to them in this case stocks and money i recommend that players have a calculator handy especially for the endgame the poor production values when compared to versions previous to it might turn some people off but id recommend putting that aside in your mind to enjoy the game for what it really is	0.0005
1636	if you only employ one creativityenhancing resource for the rest of your life make that resource the ball of whacks breakthroughs in effective creativityinspiring methods seldom occur most new methods are simply restatements or rechasings of ideas that flourished among the ancient greeks such as heraclitus a favorite source for mr von oech or before those who are familiar with the earlier work usually dont gain much from the newer approaches the ball of whacks is a happy exception heraclitus would say that you can never interact with the ball of whacks twice in the same way and he would be right the most valuable creativityenhancing methods ive experienced usually require having a team work together with one of the tasks involving using little models or prototypes of creative concepts i often find myself visiting i could have those experiences more often but seldom having the right people and time to pursue that desire them i discovered roger von oechs ball of whacks wow my frustrations are now behind me this is day 8 of working with the ball of whacks its been quite an education i can hardly wait to find out what ill learn on day 9 when you first see the ball of whacks you want quite know what to make of it open the box take out the ball the magnetic stand and the workbook then let yourself go discover what the ball is to the geometrically inclined you'll notice that its a rhombic triacontahedron try saying that 5 times fast a ballline shape with edges formed by 30 rhombi a rhombus is a foreshortened equilateral with two pairs of parallel sides that resemble scaleldown versions of an egyptian pyramid done in red the ball feels somewhat alive however it grows when you stop squeezing it as soon as you take out a piece youll discover that the ball is composed of 30 design blocks that use magnetism to adhere to one another as you take the pieces apart the ball disappears and many regular and irregular shapes occur usually thats the end for me i can never recover the original shape my rubiks cube is still a mess but the ball is easy to reform if i can do it you can too i like abstract shapes and i was pleased to find that the magnets act both to attract and repel the pieces unlike lego pieces connections can be made in all kinds of odd ways i also noticed that the pieces often reshape themselves almost organically as they choose a different connection than the one that i initiated the results are often better than my planned ones based on a suggestion in the workbook i began mixing in other objects that could be held by the design blocks the results were infinitely more interesting and rewarding than anything id ever created before i felt a great sense of peace from the experience many people had told me i would like sculpting but i lacked the confidence to try with the ball of whacks i felt like a super sculptor but without any lessons after several days of working on abstract shapes i decided to try geometric shapes these were paradoxically harder to do in many cases the magnetism sometimes fought me i had to design clever solutions to create the regularity i wanted to see i also left the ball of whacks out for people to see what others would do everyone noticed that i kept changing it without thinking revealing new designs would emerge in a few seconds i gained confidence in my creativity involving forms at the same time i was working on a number of nonphysical problems soon i began seeing metaphors in the blocks and my designs for those problems here and better solutions began to flood my mind feeling like i should read more of the workbook i now found myself intrigued by the directions for how to use the ball of whacks to expand my creative repertoire i began to see huge gaps between the methods i prefer to use gaps that i could easily explore and understand with my new appreciation of my creativity habits best of all the lessons are deeply ingrained the human nervous system is mostly made up of connections between the hands and eyes to and from the brain by both holding and seeing what i was working on the exercise connects more parts of my brain and experience i find totally different memories being stimulated especially those from the sandbox and while playing with modular toys today i went back and totally redid a book proposal im working on i was astonished to see that my mind was connecting those little blocks from the ball of whacks into new ways of assembling the pieces of the proposal i would never have expected that what does the name mean i dont think there just one meaning it obviously sounds like ball of wax which is something that warms and becomes malleable in our hands im reminded of the red wax that you peel off of edam cheese to form shapes at another level a whack in von oechs vocabulary is a way of getting you to let go of artificial assumptions in that sense this is a tool to give you lots of whacks and it does i haven't done all of the exercises yet but i plan to each of the ones ive done has been quite an eyeopener heres a tip turn off the tivo while youre watching television and use the commercial breaks to play with the ball the baseball playoffs are excellent for this purpose because you get all kinds of short delays be smart give a ball of whacks to each of your children rather than purchasing another toy youll inspire genius that will constantly fascinate you as a dad of two boys im always on the lookout for activities for us to do together something we can all enjoy and equally get into we built a da vinci catapult a stage tower did some exploring with microscopes geocaching etc with varying degrees of interest in one of the two boys dont get me wrong we all had fun but it was more for one or the other depending on the activity my older son since the day he could has loved to talk he loves to tell stories about things has done things heid like to do or things has watched if he can make a story out of something he will my younger son has since i can remember loved to play board games there are ones he likes more than others but i dont think were played a game he didnt like i recently became aware of a cardboard game by a company out of the uk named cubicle7 the game is called hobbit takes from the green dragon inn and it looked very interesting cubicle7 also makes a rpg that takes place in the same world so this game called the one ring the cards in this game can be used with that rpg if desired so its all very integrated the basic premise of the game is that the players are hobbits sitting around a table at the green dragon inn telling stories of their adventures the game takes place in rounds one for each player and they tell the story with the aid of adventure cards 75 of those each adventure card has a story aid something to inspire the story teller a title and a tokken quote along with some excellent artwork to really inspire you however the other players have similar cards called hazard cards all of those that they use to try to throw the story off things like a goblin patrol or a group of forgends this is the real magic of the game when the players interact with each other in the telling of their or each others story the goal of course is to score the most points of course but the fun isnt	0.0002

Recommendation

All applied models identified spending score and remuneration as two main drivers of changes in loyalty.

While remuneration cannot be directly influenced by the business, increasing the spending score can be triggered with promotional and marketing efforts resulting in higher loyalty score:




```r
### Predict loyalty for a customer with known attributes

# Scenario 1: low income, low spending
new_customer <- data.frame(
  remuneration = 40,
  spending_score = 40
)
predicted_loyalty <- predict(model, newdata = new_customer)
print(predicted_loyalty)
# Result = 977.6939

# Scenario 2: low income, average spending
new_customer <- data.frame(
  remuneration = 40,
  spending_score = 50
)
predicted_loyalty <- predict(model, newdata = new_customer)
print(predicted_loyalty)
# Result = 1300.814

# Scenario 3: low income, high spending
new_customer <- data.frame(
  remuneration = 40,
  spending_score = 70
)
predicted_loyalty <- predict(model, newdata = new_customer)
print(predicted_loyalty)
# Result = 1947.055
```

Recommendation: Increase customer spending to boost loyalty (1/2)

PRIORITY A customer segments:

Detached Buyers (17%)

- **Traits: High income | Very low spending | Low loyalty | Mostly aged 35–44 | Mostly highly educated**

➢ **Re-engage through premium experiences:**

- Offer exclusive previews of upcoming releases
- Provide early access to limited-edition products
- Send premium sample items (miniatures, demo packs)
- Run storytelling campaigns highlighting product quality and brand values

Steady Buyers (39%)

- **Traits: Average income, spending & loyalty | Mostly graduates | 60% females**

➢ **Grow basket size among already engaged customers:**

- Use personalized recommendations based on past purchases
- Promote limited-time bundles (e.g., game + expansion)
- Suggest cross-sell items (accessories, add-ons)
- Share educational content (how-to-play guides, strategy tips)

Recommendation: Increase customer spending to boost loyalty (2/2)

PRIORITY B customer segments:

Premium Buyers (18%)

- **Traits: High income & spending | Very high loyalty | Mostly aged 25–44 | Mostly highly educated**

➢ **Reinforce premium loyalty:**

- Provide VIP early access to new releases
- Offer exclusive loyalty-tier perks (collectibles, digital bonuses)

Budget Buyers (14%)

- **Traits: Low income & spending | Very low loyalty | > 50% aged btw. 35-54 | Mostly highly educated | >60% females**

➢ **Maintain at low cost:**

- Use automated seasonal discounts
- Offer occasional low-cost promotions (clearance sales, budget bundles)

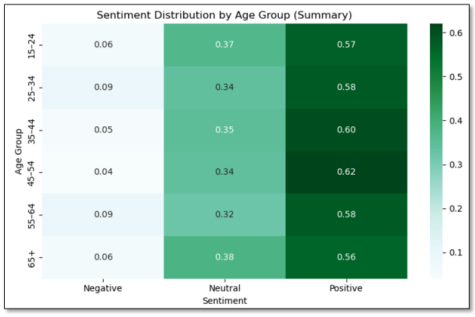
Impulsive Buyers (13%)

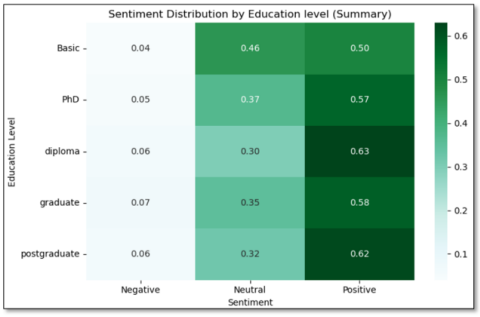
- **Traits: Low income & loyalty | High spending | Mostly 15-34 | Mostly highly educated**

➢ **Encourage Consistent Purchasing:**

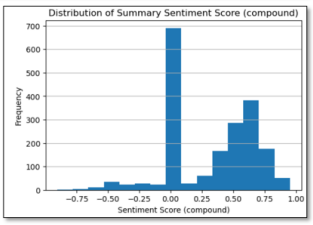
- Introduce delayed rewards (“Buy 2 this month, get 1 accessory next month”)
- Reward consistent monthly purchases (streak-based loyalty boosts)
- Use “complete your set” nudges for collectible lines

Additional insight: Overall positive product sentiment with isolated quality issues





- **Definite positive trend**, especially among age 45-54 and higher levels of education
- Negative comments mostly related to product numbers: 5429 (4x), 9119 (3x), 466 (3x), 10196 (3x), 1241 (3x) commenting product quality → **check internally**
- Limitation: most comments classified as neutral, scoring is inconsistently assigned -> **tool improvement necessary**

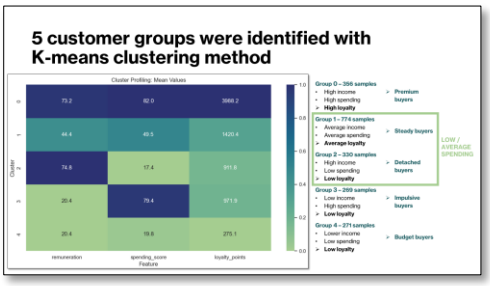


Conclusion

1. Loyalty is driven more by **behavioral factors - spending and remuneration** than demographics

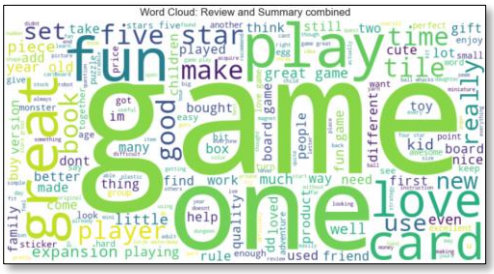
2. **Target** biggest customer groups with low/average spending
   → ***Detached buyers***
   → ***Steady buyers***

   to drive loyalty



3. If financially feasible, introduce targeted, segment-specific actions also for other customer segments

4. **Leverage sentiment insights** to guide product improvements and clearer customer messaging; supported by more refined Natural Language Processing (NLP) tooling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